PCBC H Parametric & Sequences Test (Practice B) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A baseball is hit from a height of 4 feet at an angle of 30° with an initial speed of 96 ft/s.

a) Write a set of parametric equations for the motion of the baseball. (Recall: $x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the baseball.

c) Determine how long the baseball is in the air.

d) Determine how far the baseball traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $\frac{1}{2}, -\frac{3}{4}, \frac{5}{6}, -\frac{7}{8}, \cdots$

Express the sum using summation notation. (3 points)

9. $2+\frac{3}{4}+\frac{4}{9}+\cdots+\frac{13}{144}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{18}(4-k)$

11. $\sum_{k=1}^{20}\left(3k^2+2k\right)$

12. $\sum_{k=5}^{15} 2k^3$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x-1)^7$

14. $(3x+2)^4$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Write out the sum.

E1. $\displaystyle\sum_{k=0}^{n} \frac{3^k}{k+1}$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $-1,\ \dfrac{1}{4},\ -\dfrac{1}{9},\ \dfrac{1}{16},\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Baseball: $h = 4$ ft, $\theta = 30°$, $v = 96$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (96\cos 30°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (96\sin 30°)t - 16t^2 + 4$ — *+h: starts at 4 ft*

$x_1(t) = 96\cos(30°)\cdot t,\ \ y_1(t) = 96\sin(30°)\cdot t - 16t^2 + 4$ — *what you type in*

$0 \le t \le 4$, tstep 0.01; x : 0 to 300, y : 0 to 50 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 40.000 \text{ ft}}$ — *trace to the top ($t \approx 1.500$); not Analyze, Maximum*

c) $\boxed{t \approx 3.081 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 256.161 \text{ ft}}$ — *same point, read x*

Units: feet (the speed was ft/s).

8. $\frac{1}{2}, -\frac{2}{4}, \frac{3}{5}, -\frac{4}{8}, \ldots$

top: $2n-1$, bottom: $2n$ — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = (-1)^{n+1}\frac{2n-1}{2n}}$ — *check $n = 1$: $\frac{1}{2}$*

9. $2 + \frac{3}{4} + \frac{4}{9} + \cdots + \frac{13}{144}$

$2 = \frac{2}{1}$: top $k+1$, bottom k^2 — *write 2 as $\frac{2}{1}$; tops 2, 3, 4; bottoms 1, 4, 9*

$144 = 12^2 \Rightarrow k = 12$ — *check the top: $12+1 = 13$*

$\boxed{\sum_{k=1}^{12} \frac{k+1}{k^2}}$

10.

$\sum_{k=1}^{18} 4 - \sum_{k=1}^{18} k$ — *(3) split*

$4(18) - \frac{18(19)}{2}$ — *(5) with a constant: $\sum c = cn$; (6)*

$\boxed{-99}$

11.

$3\sum_{k=1}^{20} k^2 + 2\sum_{k=1}^{20} k$ — *(2) split, (1) constants out*

$3\left[\frac{20(21)(41)}{6}\right] + 2\left[\frac{20(21)}{2}\right]$ — *(7), (6) with $n = 20$*

$\boxed{9030}$

12.

$2\sum_{k=5}^{15} k^3$ *(1)*

$2\left[\sum_{k=1}^{15} k^3 - \sum_{k=1}^{4} k^3\right]$ *(4): starts at 5, so $j=4$*

$2\left[\left(\frac{15(16)}{2}\right)^2 - \left(\frac{4(5)}{2}\right)^2\right]$ *(8)*

$\boxed{28600}$

13. $(x-1)^7$: $n=7$, row 1 7 21 35 35 21 7 1. The sign goes with its term.

$1x^7(-1)^0 + 7x^6(-1)^1 + 21x^5(-1)^2 + 35x^4(-1)^3$ *blanks first: first term's powers count down, second term's count up*

$+35x^3(-1)^4 + 21x^2(-1)^5 + 7x^1(-1)^6 + 1x^0(-1)^7$

$\boxed{x^7 - 7x^6 + 21x^5 - 35x^4 + 35x^3 - 21x^2 + 7x - 1}$ *second term negative: odd powers of -1 are negative, so signs alternate*

14. $(3x+2)^4$: $n=4$, row 1 4 6 4 1. The sign goes with its term.

$1(3x)^4(2)^0 + 4(3x)^3(2)^1 + 6(3x)^2(2)^2 + 4(3x)^1(2)^3$ *blanks first: first term's powers count down, second term's count up*

$+1(3x)^0(2)^4$

$(3x)^4 = 81x^4, \ldots$ *the whole first term gets the power*

$\boxed{81x^4 + 216x^3 + 216x^2 + 96x + 16}$ *second term positive: every term positive*

Extra practice (not scored)

E1. $\sum_{k=0}^{n} \frac{3^k}{k+1}$

$\frac{3^0}{1} + \frac{3^1}{2} + \frac{3^2}{3} + \cdots + \frac{3^n}{n+1}$ *start at $k=0$: $3^0=1$*

$\boxed{1 + \frac{3}{2} + 3 + \cdots + \frac{3^n}{n+1}}$ *don't add*

E2. $-\frac{1}{1}, \frac{2}{4}, -\frac{3}{9}, \frac{4}{16}, \ldots$

top: 1, bottom: n^2 *write 1 2 3 4 above; top and bottom separately*

$-,+,-,+ \Rightarrow (-1)^n$ *first term negative*

$\boxed{a_n = \frac{(-1)^n}{n^2}}$ *write -1 as $-\frac{1}{1}$ to see the pattern*